Ahmed Essam Mohamed

Data Engineer

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Objective

Data Engineer and Machine Learning Engineer with a strong background in data architecture, ETL processes, and predictive analytics. Passionate about designing scalable data pipelines, optimizing data storage solutions, and deploying machine learning models into production. Seeking to leverage expertise in Python, SQL, Big Data technologies (Hadoop, Spark), and cloud computing (Azure) to drive data-driven decision-making and develop intelligent solutions that enhance business efficiency.

Education

Bachelor's Degree in Biomedical Engineering – HTI University *(Sep 2016 – Jan 2022)*

- Relevant Coursework: Data Analysis, Programming (Python & MATLAB), Signal Processing, AI in Healthcare
- Graduation Project: Voice-Controlled Wheelchair *(Grade: Excellent)*

Experience and Internships

Intern – Big Data (Orange Digital Center) | Jan 2025 – Present

- Assisted in processing and analyzing large-scale datasets using Apache Spark.
- Worked with Hadoop (HDFS, MapReduce) for distributed data storage and processing.
- Developed ETL pipelines using Airflow to automate data workflows.
- Managed and optimized queries in SQL (PostgreSQL, Microsoft SQL Server) for Big Data analytics.
  Key Skills: Apache Spark, Hadoop (HDFS, MapReduce), Airflow, ETL, SQL, Big Data Analytics

Data Scientist and Machine Learning Engineer (Instant | Course) | Mar 2024 – Nov 2024

- Assisted in building predictive models and conducted data preprocessing tasks using Python and Scikit-learn.
- Performed exploratory data analysis (EDA) to uncover trends and patterns in large datasets.
- Developed dashboards and visualizations in Power BI to present data insights to stakeholders.
- Developed and fine-tuned a machine learning model that achieved 92% accuracy on the validation dataset, showcasing strong analytical and model optimization skills.
  Key Skills: Python, Scikit-learn, Machine Learning, EDA, Data Visualization, Power BI, Model Optimization

Certifications

- **Machine Learning Nanodegree – Udemy (Apr 2020) – View Certificate**
- **Microsoft SQL Server from Scratch – Udemy (Jan 2024) – View Certificate**
- **Associate Data Engineer in SQL – Data Camp (Jan 2025) – View Certificate**
- **Data Science Diploma (120 hours) – INSTANT (Nov 2024) – View Certificate**

Skills

Technical Skills

- Scripting: Python
- Data Manipulation & Preprocessing: NumPy, Pandas, SQL
- Visualization: Seaborn, Matplotlib, Power BI, Excel
- Data Engineering: ETL processes for integrating data from different sources into a single data warehouse
- Big Data Tools: Experience handling large datasets using Hadoop
- Cloud Computing: Familiarity with Azure
- Model Deployment & Production: Fast API for building APIs to deploy machine learning models

Soft Skills

- Communication
- Problem-Solving
- Time Management
- Hard Work
- Critical Thinking
- Storytelling with Data

Projects

- **Taxi Price Regression Model (Achieved 82% accuracy) – View Project**
- **Breast Cancer Analysis (Achieved 97% accuracy) – View Project**
- **Regression with Insurance Dataset (Achieved 91% accuracy) – View Project**
- **Hospital Database Analysis – View Project**
- **Bank Customers EDA – View Project**